

LUCKY SAH

Mechatronics Engineering | Robotics Hardware | Mechanical Design | Prototyping

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PROFESSIONAL SUMMARY

Mechatronics Engineering undergraduate with hands-on experience designing and prototyping robotic systems, including mobile robots, expressive robot mechanisms, and modular AI companion hardware. Skilled in CAD, mechanical-electrical integration, actuation, embedded systems, 3D printing, and fabrication. Experienced in translating concepts from parametric CAD assemblies into functional prototypes in research and product-development environments.

TECHNICAL SKILLS

CAD & Mechanical Design: Autodesk Inventor Professional, Fusion 360, SolidWorks, parametric assemblies, 3D modeling, BOM development

Robotics & Integration: Actuation systems, sensors, motor drivers, mechanical-electrical integration, ROS.

Embedded & Controls: Arduino, microcontrollers, sensor integration, PLC fundamentals, industrial automation

Fabrication: FDM/resin 3D printing, CNC machining, soldering, power tools, silicone, fiber, and polyurethane molding.

EXPERIENCE

Robotics Hardware Intern (Co-op) — Open Drive LLC

Plaipin Product Line

- Translate CAD designs into functional robotic hardware, prototypes, and integrated mechanical subsystems.
- Support mechanical design, assembly, actuation integration, and iterative prototype development for an AI companion robot platform.

Prototyping Manager — Advanced Telerobotics Research (ATR)

[ADD START DATE] – Present

Lab, Kent State University

- Lead mechanical prototyping for robotics research projects from CAD through fabrication, assembly, and functional hardware.
- Design and integrate structures, mounts, actuation components, sensors, and embedded hardware for robotic systems.
- Coordinate prototyping resources, equipment, workspace, and project support across interdisciplinary research teams.

Automation / Product Development Intern — Thoplo Machine

Pvt. Ltd. & Baliyo Ventures Pvt. Ltd. (2022–2024)

- Contributed to the design, fabrication, assembly, and testing of automated production machinery and mechanical systems.
- Supported PLC lab setup, wiring, industrial equipment assembly, machining, fiber molding, and production-oriented fabrication.

SELECTED ROBOTICS PROJECTS

Plaipin — Modular AI Companion Robot Platform

- Led mechanical CAD design of the core device and modular body configurations; developed assemblies integrating actuation, sensors, and onboard hardware.

Mini TurtleBot — Low-Cost Modular Mobile Robot | ATR Lab

- Designed the chassis and motion architecture for low-cost fabrication, modular expansion, and integration of robotic hardware.

KAMI Head — Human-Robot Interaction Module

- Designed an expressive robotic head assembly with mechanical motion systems and embedded components for human-robot interaction.

EDUCATION

Kent State University — B.S. Mechatronics Engineering, Minor in Computer Science, Kent, OH

Expected 2028

Focus: Robotics, mechanical systems, electronics, prototyping, and applied engineering design.

LEADERSHIP, COMPETITIONS & TECHNICAL ACTIVITIES

Founder & Organizer, Young Scientist Community (under RECAST - Tribhuvan University) — Led robotics education, technical workshops, STEM events, coordination, and outreach.

Selected competitions/conferences: World Robot Olympiad (Thailand), FIRST Global Challenge (Switzerland), Asian Development Bank Hackathon, Robo-Battle Bots, Robo Soccer, Founder's Inc (San Francisco), and Asian Pacific Conference of Young Scientists.